

Linux Class 11 - Practice Task

Practising ps, kill and find

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This is my practice for the commands we went through in class 11. I made a folder called practice11 and put a few different things inside it.

What I made	Why I needed it
notes.txt	A normal file with something written in it.
empty.txt	An empty file, to try find -empty.
old.txt	A file made to look 2 hours old, to try find -mmin +15.
docs (folder)	A folder with a file inside, so find has something to go into.
emptydir (folder)	An empty folder, to try find -empty -type d.
link.txt	A link pointing to notes.txt, to try find -type l.

The only one that needed a trick was old.txt. Everything I make is new, so there would be nothing for find -mmin +15 to show. I used touch with a date to make one file look two hours old:

```
touch -d "2 hours ago" old.txt
```

1. ps - seeing what is running

I started with ps on its own. It gave only two lines, my bash and the ps command itself, because plain ps shows only the processes of the terminal I am sitting in.

```
ps
```

Then I added -f, which stands for full. The same two processes came, but this time with the UID, PID, PPID and the time they started.

```
ps -f
```

To have something more interesting to look at, I started a process that does nothing for 300 seconds. The & at the end puts it in the background so that I get my prompt back straight away:

```
sleep 300 &
```

```
ps -f
```

Now ps -f showed three processes instead of two, and the sleep 300 was there in the list with its own PID.

```
Shiva@DESKTOP-MRCVEKT: ~/practice11
Shiva@DESKTOP-MRCVEKT:~/practice11$ ps
  PID TTY          TIME CMD
  6248 pts/0        00:00:00 bash
  6447 pts/0        00:00:00 ps
Shiva@DESKTOP-MRCVEKT:~/practice11$ ps -f
UID          PID    PPID  C STIME TTY          TIME CMD
Shiva        6248    6241  0 10:19 pts/0        00:00:00 -bash
Shiva        6457    6248  99 10:20 pts/0        00:00:00 ps -f
Shiva@DESKTOP-MRCVEKT:~/practice11$ sleep 300 &
[1] 6467
Shiva@DESKTOP-MRCVEKT:~/practice11$ ps -f
UID          PID    PPID  C STIME TTY          TIME CMD
Shiva        6248    6241  0 10:19 pts/0        00:00:00 -bash
Shiva        6467    6248  22 10:20 pts/0        00:00:00 sleep 300
Shiva        6477    6248  99 10:20 pts/0        00:00:00 ps -f
Shiva@DESKTOP-MRCVEKT:~/practice11$
```

Screenshot 1: ps, then ps -f, then a background process appearing in the list.

The PID column is the important one. That number is what every other command needs when I want to do something to a process.

2. ps -ef - every process on the system

ps -ef is the one my notes say shows the whole system and not just my terminal. When I ran it the output was far too long for the screen, so I sent it into a file first and then counted the lines and looked at the top of it:

```
ps -ef > all.txt
wc -l all.txt
head -8 all.txt
```

It came to 35 lines, against the 3 that plain ps gave me. And the names in the first column are root, not Shiva, so these are the system's own processes, like init and the systemd ones, which is exactly what the e in -ef means.

```
Shiva@DESKTOP-MRCVEKT: ~/practice11
Shiva@DESKTOP-MRCVEKT:~/practice11$ ps -ef > all.txt
Shiva@DESKTOP-MRCVEKT:~/practice11$ wc -l all.txt
35 all.txt
Shiva@DESKTOP-MRCVEKT:~/practice11$ head -8 all.txt
UID          PID    PPID  C STIME TTY          TIME CMD
root          1         0  0 Oct04 ?        00:00:04 /sbin/init
root          2         1  0 Oct04 hvcd0 00:00:00 /init
root          3         2  0 Oct04 hvcd0 00:00:00 init-watcher
root          9         2  0 Oct04 hvcd0 00:00:00 plan9 --control-socket 7 --log-level 4 -
-server-fd 8 --pipe-fd 10 --log-truncate
root         52         1  0 Oct04 ?        00:00:03 /usr/lib/systemd/systemd-journald
systemd+     93         1  0 Oct04 ?        00:00:00 /usr/lib/systemd/systemd-resolved
root         97         1  0 Oct04 ?        00:00:04 /usr/lib/systemd/systemd-udev
Shiva@DESKTOP-MRCVEKT:~/practice11$
```

Screenshot 2: ps -ef has 35 lines, and they belong to root, not to me.

My notes also say that ps is a snapshot. I understand that now as meaning it shows what was running at the moment I pressed Enter and then it stops, while top keeps redrawing the list until I press q.

3. kill - stopping a process

To practise this I started a sleep again, found its PID with ps -f, and then stopped it:

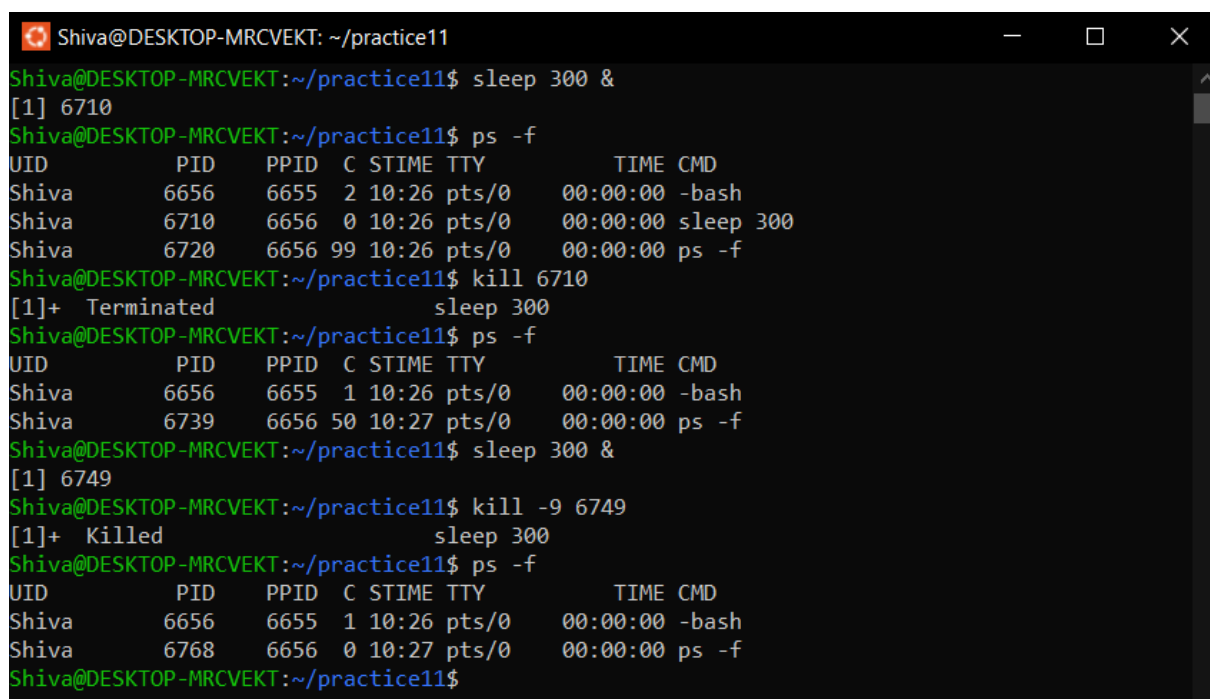
```
sleep 300 &
ps -f
kill 6710
```

The shell answered Terminated, and when I ran ps -f again the sleep was gone from the list. Then I started another one and used the forceful form on it:

```
kill -9 6749
```

This time the shell answered Killed instead of Terminated. That was the part I did not expect, and it is the easiest way to see the difference between the two:

- kill asks the process to stop, so it gets a chance to finish what it is doing. The shell says Terminated.
- kill -9 does not ask, it stops it at once. The shell says Killed.

A screenshot of a terminal window titled 'Shiva@DESKTOP-MRCVEKT: ~/practice11'. The terminal shows the following sequence of commands and outputs:
1. `sleep 300 &` followed by `[1] 6710`.
2. `ps -f` showing a table of processes including the `sleep 300` process with PID 6710.
3. `kill 6710` followed by `[1]+ Terminated sleep 300`.
4. `ps -f` showing the process list after the first kill.
5. `sleep 300 &` followed by `[1] 6749`.
6. `kill -9 6749` followed by `[1]+ Killed sleep 300`.
7. `ps -f` showing the process list after the second kill.
The terminal output for `ps -f` is as follows:

UID	PID	PPID	C	STIME	TTY	TIME	CMD
Shiva	6656	6655	2	10:26	pts/0	00:00:00	-bash
Shiva	6710	6656	0	10:26	pts/0	00:00:00	sleep 300
Shiva	6720	6656	99	10:26	pts/0	00:00:00	ps -f

UID	PID	PPID	C	STIME	TTY	TIME	CMD
Shiva	6656	6655	1	10:26	pts/0	00:00:00	-bash
Shiva	6739	6656	50	10:27	pts/0	00:00:00	ps -f

UID	PID	PPID	C	STIME	TTY	TIME	CMD
Shiva	6656	6655	1	10:26	pts/0	00:00:00	-bash
Shiva	6768	6656	0	10:27	pts/0	00:00:00	ps -f

Screenshot 3: kill says Terminated, kill -9 says Killed, and ps -f proves both worked.

So I should try plain kill first, and only use -9 if the process refuses to stop, because -9 gives it no time to save anything.

4. find - listing by type

I started with find on its own, with a dot for the folder I am in:

```
find .
```

It listed everything together, the files, the folders and the link, and it also went inside the docs folder on its own without me asking. Then I narrowed it down with -type:

```
find . -type f
find . -type d
find . -type l
```

f gave only the files, d gave only the folders, and l gave only link.txt. I noticed that -type d also prints the dot at the top, because the folder I am standing in counts as a folder too.

```
Shiva@DESKTOP-MRCVEKT: ~/practice11
Shiva@DESKTOP-MRCVEKT:~/practice11$ find .
.
./emptydir
./empty.txt
./notes.txt
./all.txt
./docs
./docs/menu.txt
./old.txt
./link.txt
Shiva@DESKTOP-MRCVEKT:~/practice11$ find . -type f
./empty.txt
./notes.txt
./all.txt
./docs/menu.txt
./old.txt
Shiva@DESKTOP-MRCVEKT:~/practice11$ find . -type d
.
./emptydir
./docs
Shiva@DESKTOP-MRCVEKT:~/practice11$ find . -type l
./link.txt
Shiva@DESKTOP-MRCVEKT:~/practice11$
```

Screenshot 4: everything, then only files, then only folders, then only links.

5. find - empty things and permissions

Next I tried the ones that look at what is in a file rather than what kind of thing it is:

```
find . -empty -type f
find . -empty -type d
```

The first gave empty.txt and old.txt, because old.txt is empty as well as old. The second gave emptydir.

Then the permission ones. Earlier I had given notes.txt full permission with chmod 777, so there was something to find:

```
find . -perm 777
find . -perm 777 -type f
```

The first one gave two things, notes.txt and link.txt. That surprised me at first, but my link has 777 permission as well, because a link always shows lrwxrwxrwx whatever the file behind it is. Adding -type f took the link out and left only notes.txt.

```
Shiva@DESKTOP-MRCVEKT: ~/practice11
Shiva@DESKTOP-MRCVEKT:~/practice11$ find . -empty -type f
./empty.txt
./old.txt
Shiva@DESKTOP-MRCVEKT:~/practice11$ find . -empty -type d
./emptydir
Shiva@DESKTOP-MRCVEKT:~/practice11$ find . -perm 777
./notes.txt
./link.txt
Shiva@DESKTOP-MRCVEKT:~/practice11$ find . -perm 777 -type f
./notes.txt
Shiva@DESKTOP-MRCVEKT:~/practice11$
```

Screenshot 5: empty files, empty folders, and the two things with 777 permission.

6. find - by time

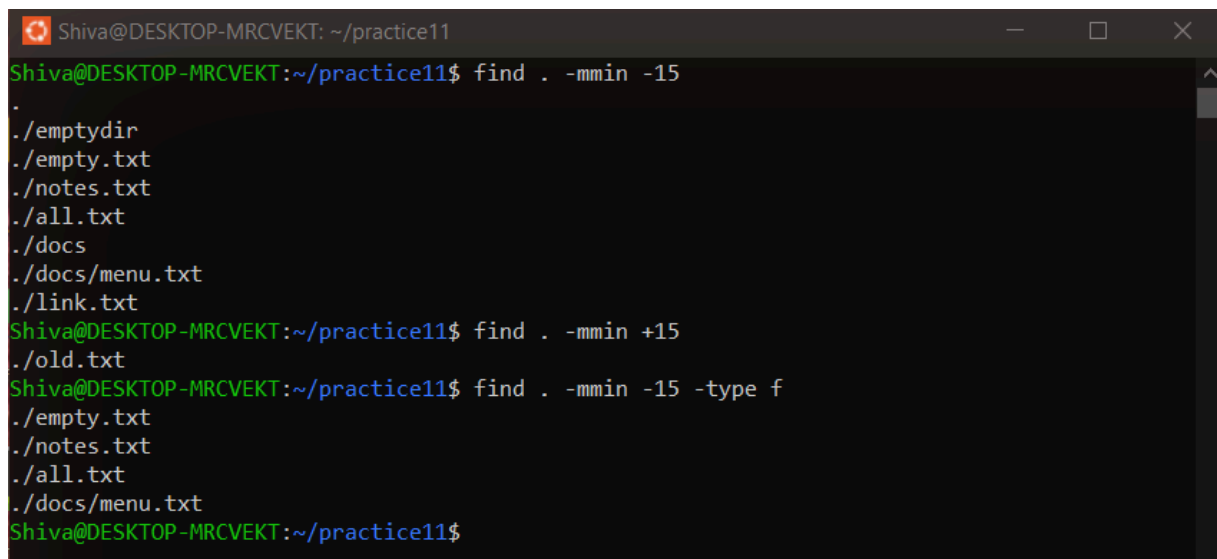
Last I practised the time options. My notes list three of them, and the sign in front of the number is what decides the meaning:

```
find . -mmin -15
```

```
find . -mmin +15
```

```
find . -mmin -15 -type f
```

The minus gave everything I had made that afternoon. The plus gave only old.txt, the one I had backdated, which proves the plus means older than 15 minutes and not within 15 minutes. The third one is the two options joined together, so only the recent files and not the folders.



```
Shiva@DESKTOP-MRCVEKT: ~/practice11
Shiva@DESKTOP-MRCVEKT:~/practice11$ find . -mmin -15
.
./emptydir
./empty.txt
./notes.txt
./all.txt
./docs
./docs/menu.txt
./link.txt
Shiva@DESKTOP-MRCVEKT:~/practice11$ find . -mmin +15
./old.txt
Shiva@DESKTOP-MRCVEKT:~/practice11$ find . -mmin -15 -type f
./empty.txt
./notes.txt
./all.txt
./docs/menu.txt
Shiva@DESKTOP-MRCVEKT:~/practice11$
```

Screenshot 6: minus 15 is the recent things, plus 15 is only the old file.

The three letters are for three different times. -mmin is when the file was last changed, -amin is when it was last read, and -cmin is when the details of the file changed, like its name or permission. A plain number with no sign means exactly that minute, which almost never matches anything.

Things I noticed while practising

- Plain ps only knows about my own terminal. ps -ef is the one that sees the whole machine.
- Nothing can be killed without the PID, so ps always comes before kill.
- kill says Terminated, kill -9 says Killed. That is how I can tell which one actually did the job.
- find goes into sub folders by itself. I did not have to ask for it.
- A link always has 777 permission, so -perm 777 finds links even when I was looking for files.
- With -mmin, the minus and plus signs are the whole meaning. Getting them the wrong way round gives the opposite answer.

Commands I practised

Command	What it does
ps	The processes of my own terminal, short form.
ps -f	The same, with full details and the PID.

Command	What it does
ps -ef	Every process of every user on the system.
sleep 300 &	Starts a process in the background to practise on.
kill 6710	Asks the process with that id to stop.
kill -9 6749	Forces it to stop.
find .	Everything in this folder and the folders inside it.
find . -type f	Only the files.
find . -type d	Only the folders.
find . -type l	Only the links.
find . -empty -type f	Only the empty files.
find . -empty -type d	Only the empty folders.
find . -perm 777	Everything with 777 permission.
find . -mmin -15	Everything changed in the last 15 minutes.
find . -mmin +15	Everything changed more than 15 minutes ago.